

# Minimal Diagnostic Principles of Life: Criterion-Ablation and Observable Surrogate Selection in a Digital Ecosystem

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## Abstract

We present a testable integration implementing all seven textbook biological criteria for life—cellular organization, metabolism, homeostasis, growth, reproduction, response to stimuli, and evolution—as functionally interdependent computational processes within a single artificial life system. Existing systems implement at most a subset of these criteria, often as independent modules or static proxies that can be removed without measurable system degradation. Our hybrid swarm-organism architecture implements each criterion as a dynamic process satisfying three conditions—sustained resource consumption, measurable degradation upon removal, and feedback coupling with other criteria—which we term functional analogy. A criterion-ablation experiment ( $n=30$  per condition, seeds held out from calibration) demonstrates that disabling any single criterion causes statistically significant population decline (Mann-Whitney  $U$ , Holm-Bonferroni corrected, all  $p \leq 0.004728$ ), with Cliff’s  $\delta$  ranging from 0.39 to 1.00. Pairwise ablations reveal sub-additive interactions consistent with shared failure pathways, confirming that criteria damage overlapping subsystems rather than operating independently. A proxy control comparison shows that metabolism implementations of differing complexity produce qualitatively distinct population dynamics, ruling out tautological criterion definitions. The strongest single-ablation effects arise from disabling reproduction ( $\Delta\%=-91.1$ ), response to stimuli ( $\Delta\%=-88.5$ ), and metabolism ( $\Delta\%=-86.8$ ), confirming that these criteria function as necessary, interdependent components of organismal viability rather than decorative labels. A Phase 1 criterion-reduction analysis (30 ablation conditions, 600 simulation runs; full vs. all-off Cohen’s  $d=10.68$ ) identifies six stable non-collinear surrogates via VIF screening and stability selection (Meinshausen and Bühlmann, 2010); directional Elastic Net evidence ranks metabolism, response, and homeostasis as the strongest criterion contributors. A powered Phase 2 expansion (200 conditions across 5 environment regimes, 14,000 runs) with LASSO/Elastic Net stability selection confirms 11 of 12 candidate surrogates (all stability frequencies  $\geq 0.98$ ), achieving validation  $R^2 = 0.955$  and out-of-regime test  $R^2 = 0.950$  (permutation  $p < 0.001$ ), exceeding a single-predictor baseline ( $R^2 = 0.921$ ).

Submission type: Full Paper

Data/Code available at: <https://anonymous.4open.science/>

## Introduction

What distinguishes a living system from a merely complex one? Biology textbooks identify seven criteria—cellular organization, metabolism, homeostasis, growth and development, reproduction, response to stimuli, and evolution (Urry et al., 2020)—one of several such lists in biology education, which we adopt as a working set—but artificial life (ALife) research has struggled to integrate all seven into a single computational system. This choice is pragmatic rather than exclusive: we use the seven criteria as an operational bridge across broader frameworks (NASA’s Darwinian criterion, autopoietic closure, and autonomy-centered minimal-life definitions), then test this bridge empirically via ablation. Most existing platforms implement a subset: evolutionary dynamics without metabolism (Ofria and Wilke, 2004), pattern formation without reproduction (Chan, 2019), or boundary self-organization without evolution (Plantec et al., 2023). Where criteria are nominally present, they often function as simplified proxies—static parameters or independent modules whose removal has no measurable effect on system behavior.

We argue that a meaningful computational model of life requires more than feature checklists. Each criterion must function as a functional analogy of its biological counterpart, satisfying three conditions:

1. Dynamic process: the criterion requires sustained resource consumption at every timestep, not a static lookup.
2. Measurable degradation: ablating the criterion causes statistically significant decline in organism viability.

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3. Feedback coupling: the criterion forms at least one feedback loop with another criterion, precluding independent-module implementations.

This definition is system-agnostic: the three conditions test whether a subsystem fulfills the functional role of a life criterion, not whether it structurally resembles its biological counterpart. The same conditions can be applied to any ALife system with observable state variables and ablation toggles.

This definition operationalizes the intuition behind autopoiesis (Maturana and Varela, 1980), the chemoton model’s integration of metabolic, boundary, and information subsystems (Gánti, 2003), and minimal-life frameworks (Ruiz-Mirazo et al., 2004) in a form amenable to experimental falsification.

We adopt a weak ALife stance: our system is a functional model of life, not a claim that the organisms are alive. The contribution is methodological—a two-stage design: Stage 1 integrates all seven criteria and verifies functional necessity via controlled ablation; Stage 2 applies criterion reduction (identifying which criteria contribute most to life-likeness) and surrogate selection (finding minimal observable measurements that predict life-likeness without running the full system) to the resulting 30-condition sweep.

This paper contributes: (1) a hybrid swarm-organism architecture integrating all seven criteria as functionally interdependent processes; (2) an operational functional analogy definition with three falsifiable conditions and a criterion-ablation methodology; (3) pairwise ablation and proxy control experiments providing evidence against tautological definitions; and (4) Phase 2 powered surrogate analysis (200 conditions, 5 regimes, 14,000 runs) achieving validation  $R^2 = 0.955$  and out-of-regime generalization  $R^2 = 0.950$ . The key contribution is a falsifiable experimental framework for testing functional necessity and interaction of life criteria in any ALife system.

## Related Work

Table 1 surveys eight representative ALife systems across the seven criteria. Tierra (Ray, 1991) and Avida (Ofria and Wilke, 2004) achieve strong evolution but lack spatial organization, metabolism, and homeostasis. Polyworld (Yaeger, 1994) and Geb (Channon, 2001) integrate evolution with neural-network controllers and physical embodiment, but lack explicit metabolic or homeostatic subsystems. Stringmol (Hickinbotham et al., 2011) implements chemistry-based metabolism but omits homeostasis and spatial organization. Beer’s Game-of-Life autopoiesis model (Beer, 2004) provides rigorous cellular organization but lacks evolution and reproduction. Lenia (Chan, 2019), Flow-Lenia (Plantec

Table 1: Criterion comparison (five-level rubric; bold  $\geq 4$ ). Abl.=ablation-testable (per-criterion toggle enabling controlled removal). Rubric definitions in §S4 (Table 6).

| System                                | CO | Me | Ho | Gr | Re | Rs | Ev | $\Sigma$ | Abl. |
|---------------------------------------|----|----|----|----|----|----|----|----------|------|
| Tierra (Ray, 1991)                    | 1  | 1  | 1  | 1  | 3  | 2  | 4  | 13       | –    |
| Avida (Ofria and Wilke, 2004)         | 1  | 1  | 1  | 1  | 3  | 3  | 4  | 14       | –    |
| Polyworld (Yaeger, 1994)              | 3  | 2  | 1  | 2  | 4  | 4  | 4  | 20       | –    |
| Geb (Channon, 2001)                   | 2  | 1  | 1  | 2  | 3  | 3  | 4  | 16       | –    |
| Stringmol (Hickinbotham et al., 2011) | 2  | 3  | 1  | 1  | 3  | 1  | 3  | 14       | –    |
| Beer ’04 (Beer, 2004)                 | 4  | 2  | 2  | 1  | 1  | 2  | 1  | 13       | –    |
| ALIEN (Heinemann, 2008)               | 4  | 3  | 2  | 3  | 4  | 4  | 4  | 24       | –    |
| Ours <sup>†</sup>                     | 4  | 4  | 4  | 4  | 4  | 4  | 3  | 27       | ✓    |

<sup>†</sup>Self-assessed. CO=Cell.Org, Me=Metab, Ho=Homeo, Gr=Growth, Re=Reprod, Rs=Response, Ev=Evol.

et al., 2023), ALIEN (Heinemann, 2008), and Coralai (Barbieux and Canaan, 2024) each cover additional subsets. Critically, none of these systems provides per-criterion ablation toggles enabling controlled removal experiments (Table 1, Abl. column); no existing system combines multi-step metabolism with active homeostatic regulation in an ablation-testable framework. Our framework operationalizes autopoiesis (Maturana and Varela, 1980; McMullin, 2004), the chemoton model (Gánti, 2003), and NASA’s two-criterion definition (Joyce, 1994; Ruiz-Mirazo et al., 2004); open-ended evolution metrics (Bedau et al., 2000; Taylor et al., 2016) complement our criterion-ablation approach.

We acknowledge that Table 1 relies on self-assessment for our system; scores for other systems are our best reading of published descriptions. Independent cross-validation by system developers would strengthen these comparisons.

## System Design

The system implements a hybrid two-layer architecture (Figure 1). The outer layer is a continuous toroidal 2D environment (100×100 world units) containing a diffusing resource field. The inner layer consists of 10–50 organisms, each composed of 10–50 swarm agents that collectively maintain the organism’s spatial boundary.

## Seven Criteria

Table 2 maps each biological criterion to its computational implementation, ablation toggle, and feedback partners.

Full architecture, genome encoding (256-float vector organized into 7 segments; see §S4), neural-network specification (8→16→4 tanh; 212 weights), and parameter values appear in §S4.

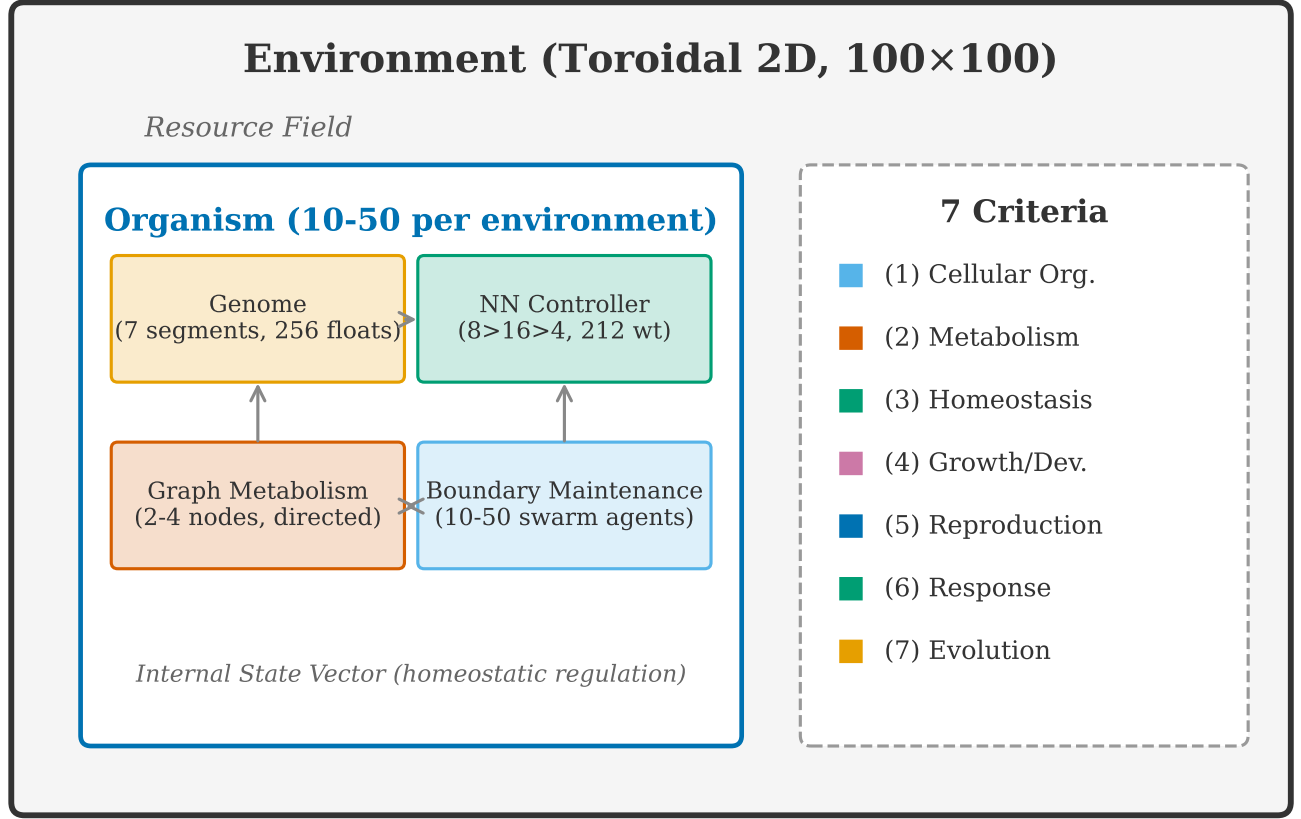


Figure 1: Two-layer architecture. Each organism comprises swarm agents maintaining a spatial boundary, a neural-network controller, a graph-based metabolic network, and a variable-length genome encoding all seven criteria. Organisms inhabit a continuous toroidal environment with a diffusing resource field.

### Criterion-Ablation Experiment

This experiment tests whether each of the seven criteria is functionally necessary for organism viability, as predicted by the functional-analogy framework.

#### Protocol

For each of the seven criteria, we disable it while keeping all others active and compare against the fully enabled baseline (8 conditions total). A mid-run extension (ablation onset at step 1000) tests continuous necessity (§S1).

#### Outcome Measures

An organism dies when  $b < 0.1$ ,  $e \leq 0.0$ , or age  $> 20,000$  steps. The primary dependent variable is  $N_T$  (alive count at  $T=2000$ ); secondary DVs are alive-count AUC and median organism lifespan. Metabolism, homeostasis, cellular organization, and response to stimuli are individual-level criteria; reproduction and evolution are population-level. Criterion-orthogonal controls (spatial

cohesion, median lifespan) are reported in §S3 (Figure 11).

#### Data Separation

Seeds 0–99 were used for calibration (threshold tuning); seeds 100–129 ( $n=30$ ) are the held-out test set for all reported results. No ablation-condition outcomes were observed during calibration; thresholds targeted stable baseline populations only. All final experiments use the Graph metabolism engine.

**Parameters.** Each simulation runs for 2000 timesteps with population sampled every 50 on a 100×100 toroidal grid with 30 initial organisms (25 swarm agents each); resources regenerate at 0.01/cell/step; ablation toggles skip disabled processes without altering event ordering. Importantly, skipping a process does not redistribute saved energy or computation to other processes—the ablation simply removes the process’s contribution, preserving the baseline resource budget. Run manifests (\*\_manifest.json)

Table 2: Mechanism specification: state variables, ablation operators, coupling pathways, and failure modes for each criterion. All criteria satisfy the three functional-analogy conditions (dynamic process, measurable degradation, feedback coupling).

| Criterion  | State Vars                                        | Ablation                | Coupling                                                               | Failure                              |
|------------|---------------------------------------------------|-------------------------|------------------------------------------------------------------------|--------------------------------------|
| Cell. Org. | $b \in [0, 1]$                                    | Skip repair; decay only | $e \rightarrow \text{repair rate} \rightarrow b$                       | $b < 0.1 \rightarrow \text{death}$   |
| Metab.     | $e, r, w$ ; graph                                 | Freeze $(e, r, w)$      | throughput $\rightarrow e \rightarrow \text{bdry repair}$              | Energy depletion                     |
| Homeo.     | $\mathbf{p}$<br>$\mathbf{s} \in \mathbb{R}^3$     | Skip NN delta; decay    | $s_0 \rightarrow \text{repair efficacy}$                               | State drift                          |
| Growth     | $m \in [0, 1]$ ;<br>$\mathbf{d} \in \mathbb{R}^8$ | Freeze $m=0$            | $m \rightarrow \text{bdry repair, sensing, metab. eff., reprod. gate}$ | Reduced repair + sensing; no reprod. |
| Reprod.    | Pop. event                                        | Skip division check     | energy cost $\rightarrow \text{parent } e$                             | No replacement                       |
| Response   | $\mathbf{v}$                                      | Skip velocity delta     | movement $\rightarrow \text{resource} \rightarrow r$                   | Starvation                           |
| Evolution  | genome $\mathbf{g}$                               | Copy w/o mutation       | $\mathbf{g}$ variation $\rightarrow \text{all}$                        | No adaptation                        |

Table 3: Criterion-ablation results ( $n=30$  per condition). Normal baseline mean: 348.9 (median: 349.5, IQR: 335.0–363.5).  $\delta$ =Cliff’s delta [bootstrap 95% CI]. Seven one-sided tests, Holm-Bonferroni corrected at  $\alpha = 0.05$ . \*\*\* $p < 0.001$ , \* $p < 0.05$ .

| Condition   | Mean  | $\Delta\%$ | $\delta$ [95% CI] | $p_{\text{corr}}$ | Sig. |
|-------------|-------|------------|-------------------|-------------------|------|
| No Reprod.  | 31.2  | −91.1      | 1.00 [1.00, 1.00] | <.001             | ***  |
| No Response | 40.0  | −88.5      | 1.00 [1.00, 1.00] | <.001             | ***  |
| No Metab.   | 46.1  | −86.8      | 1.00 [1.00, 1.00] | <.001             | ***  |
| No Homeo.   | 169.0 | −51.5      | 1.00 [1.00, 1.00] | <.001             | ***  |
| No Growth   | 188.8 | −45.9      | 1.00 [1.00, 1.00] | <.001             | ***  |
| No Boundary | 226.7 | −35.0      | 0.99 [0.98, 1.00] | <.001             | ***  |
| No Evol.    | 321.8 | −7.8       | 0.39 [0.11, 0.64] | .0047             | *    |

lock parameters to reported results (registry: docs/research/result\_manifest\_bindings.json).

### Statistical Design

We test  $H_1 : N_{\text{normal}} > N_{\text{ablated}}$  using the Mann-Whitney  $U$  test (Mann and Whitney, 1947) with Holm-Bonferroni correction (Holm, 1979) for seven simultaneous comparisons ( $\alpha = 0.05$ ). Effect sizes are Cliff’s  $\delta$  (Cliff, 1993) and (for evolution-strengthening) Cohen’s  $d$ , each with percentile bootstrap 95% CIs (2,000 resamples). The unit of replication is a seed;  $n=30$  uses seeds 100–129; outcomes are  $N_T$  at  $T=2000$  (primary) and alive-count AUC (secondary).

### Results

All seven criterion ablations produce statistically significant population decline compared to the normal baseline (Table 3).

Three ablations cause near-total collapse (>86%): reproduction, response, and metabolism—the core viabil-

ity loop (Figure 2). The key takeaway from Figure 2 is that the top three ablations converge to near-zero populations within 500 steps, while evolution ablation remains close to baseline, confirming a clear effect hierarchy. Metabolic ablation collapses within 200 steps; reproduction ablation declines progressively; evolution ablation ( $\delta=0.39$ ) shows only modest impact, consistent with adaptation at longer timescales. Reproduction-ablated organisms live longer individually (median 751 vs. 242 steps) despite population collapse, confirming absent replacement rather than individual fragility (per-seed distributions in Figure 12, §S3).

Quantitative coupling evidence. Granger-style lagged F-tests (lags 1–5, within-seed Holm correction) show temporal precedence consistent with all three hypothesized pathways ( $p_{\text{corr}} < 0.001$ ); energy and boundary integrity are strongly anti-correlated at lag 0 ( $r = -0.78$ ,  $p < 10^{-8}$ ). Intervention effects confirm non-modularity: metabolism ablation raises boundary by 41% (no repair energy) while response ablation reduces energy by 80% (organisms cannot reach resources); homeostasis ablation produces the largest internal-state change (−72%). Full coupling graph and intervention table appear in §S3 (Figure 13, Table 5).

### Proxy Control Comparison

Three metabolism implementations on the same seeds ( $n=30$ )—Counter (single-step, no waste), Toy (single-step with waste), Graph (multi-step network)—all satisfy “dynamic resource consumption” yet produce qualitatively different dynamics (Figure 3): population size differs by 27% and genome diversity varies non-monotonically, confirming that the specific implementation shapes system behavior, providing evidence against

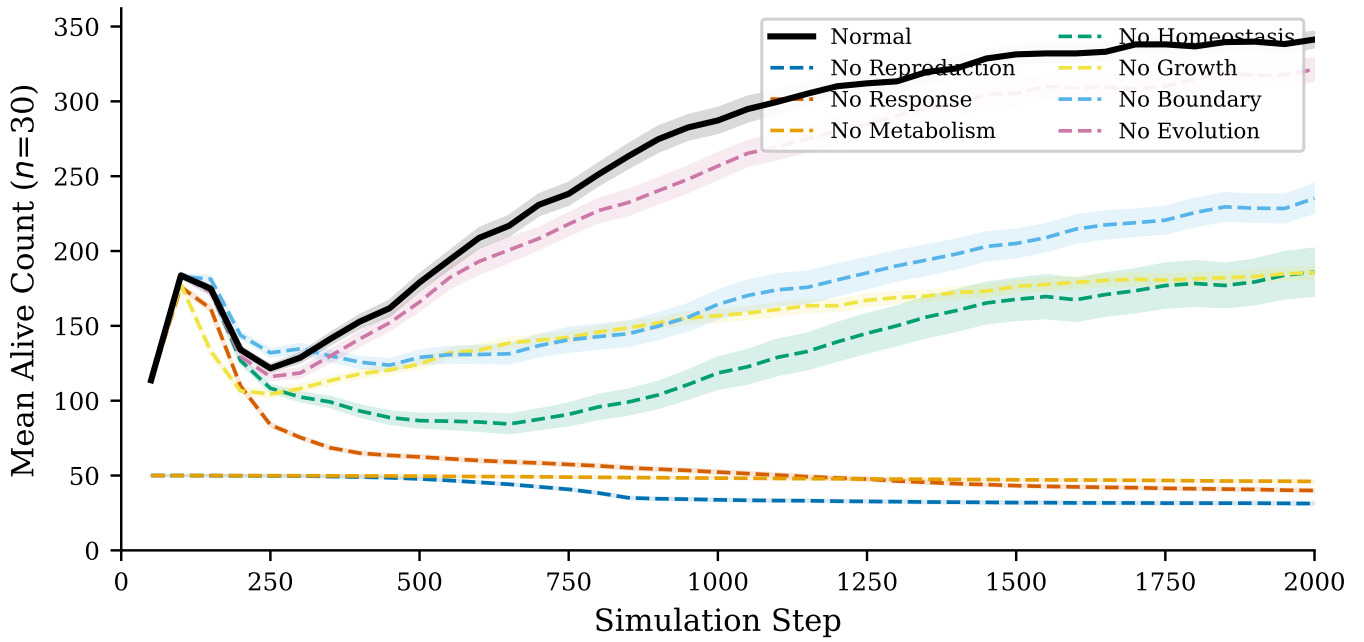


Figure 2: Population dynamics under criterion ablation. Lines show mean alive count across 30 seeds (100–129); shaded bands show  $\pm 1$  SEM. Normal baseline (thick black) stabilizes near 349 organisms by step 1900. Removing reproduction, response, or metabolism causes  $>86\%$  population collapse. Evolution ablation shows a modest 8% decline (Cliff’s  $\delta=0.39$ ), consistent with optimization rather than short-term survival necessity.

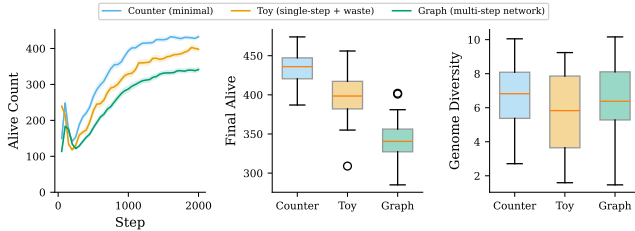


Figure 3: Proxy control comparison. Three metabolism implementations of increasing complexity on the same seeds ( $n=30$ ). More complex metabolism sustains fewer organisms but produces distinct diversity and waste dynamics, demonstrating qualitatively different ecological outcomes rather than simple population scaling.

tautological definitions. Importantly, the direction of ablation effects (metabolism removal causes population decline) is consistent across all three implementations, confirming that the qualitative conclusions are not implementation-specific.

Pairwise ablations (15 combinations; 6 reported in Table 7) show uniformly sub-additive synergy (bootstrap 95% CI excludes zero for 5/6 pairs), confirming shared failure pathways rather than independent kill switches. The (metabolism, homeostasis) pair exemplifies convergent boundary failure: both routes reduce

energy-funded boundary repair, so  $\Delta_{AB}=298.1$  falls below the additive expectation 450.7 (ratio 0.66). Full synergy formula, per-pair analysis, and the growth–reproduction interaction appear in §S5.

Evolution at longer timescales. At 10,000 steps the evolution effect grows to  $d=1.42$  ( $\delta=0.66$ ,  $p < 0.001$ ), confirming adaptation accumulates across generations; genome-drift divergence provides primary directional evidence ( $\delta=1.00$ , Figure 18A); cyclic-recovery differences are mixed ( $p=0.341$ ). See §S6 for full results and generation-stratified physiology.

Graded ablation ( $\{1.0, 0.75, 0.5, 0.25, 0.0\}$  efficiency;  $n=30$ ) shows a monotonic dose-response (Jonckheere-Terpstra,  $p < 0.001$ ), confirming metabolic contribution is quantitatively proportional rather than a binary switch. A state-neutral sham control ( $n=30$ ) yields no significant difference ( $p > 0.05$ ), validating that effects are functional. Full graded and cyclic environment results appear in §S7.

See Supplementary Material for mid-run extensions (§S1), ecology stressor protocol (§S2), phenotype clustering validation (§S3), pairwise ablation analysis (§S5), evolution timescale extension (§S6), full statistical results (§), failure mode analysis (§), fitness landscape evidence (§), threshold sensitivity and boundary topology robustness (§),

Table 4: Surrogate Analysis Summary (Phase 1 and Phase 2)

| Analysis                              | Finding                 | Value      |
|---------------------------------------|-------------------------|------------|
| Phase 1 (30 conditions, 600 runs)     |                         |            |
| PCA                                   | Comp. $\geq 90\%$ var.  | 4 (93%)    |
| Stability (Enet)                      | Top criterion           | Met. 0.678 |
| Stability (Enet)                      | Bottom criterion        | Evo. 0.374 |
| Crit. Pareto                          | $R^2$ direction         | All neg.   |
| Cohen's $d$                           | full vs. all-off        | 10.68      |
| Phase 2 (200 conditions, 14,000 runs) |                         |            |
| Stability sel.                        | Stable features         | 11/12      |
| Train $R^2$                           | Permutation $p < 0.001$ | 0.954      |
| Validation $R^2$                      | Seen regimes, new seeds | 0.955      |
| Test $R^2$                            | Unseen regimes          | 0.950      |
| Baseline $R^2$                        | alive_auc only          | 0.921      |

graded/cyclic experiments (§S11), NN controller sensitivity (§S12), counter/toy engine comparison (§S13), and growth/reproduction separability (§S14).

## Criterion Reduction Analysis

### Protocol

A 30-condition ablation sweep (24 training, 6 held-out; 20 seeds each; 600 runs total) tests criterion-reduction feasibility. Conditions cover: the full 7-criteria baseline, the all-criteria-off control, 7 single-criterion drops, and 15 pairwise drops; 6 evolution-ablation conditions are held out. Decision thresholds are pre-registered in docs/research/decision\_rules.md before held-out conditions are observed.

### Stability Selection

LASSO (Tibshirani, 1996) and Elastic Net (Enet) bootstrapped stability scores (Meinshausen and Bühlmann, 2010) (500 subsamples, threshold 0.60) are computed for each criterion across 7 performance metrics. No criterion clears 0.60 under LASSO (maximum: metabolism=0.53). Enet provides directional evidence: metabolism (0.678), response (0.641), and homeostasis (0.630) rank highest; evolution (0.374) and growth (0.492) rank lowest (Figure 4). This ordering is directionally consistent with single-ablation effect sizes (Table 3).

### PCA Structure

PCA of the 30-condition performance matrix (7 criteria  $\times$  7 metrics) requires 4 components for  $\geq 90\%$  variance (PC1=38%, PC2=31%, PC3=16%, PC4=8%; 93% cumulative; PC1+PC2=69%). The PC1-PC2 biplot (Figure 5, left) shows the full-criteria condition and the all-off control at opposite poles; pairwise ablations cluster near all-off, consistent with multiplicative criterion interdependence (Table 7).

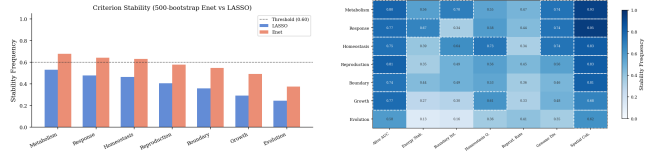


Figure 4: Criterion stability scores (left; dashed=0.60 threshold; Enet ranks metabolism>response>homeostasis highest, evolution lowest) and stability heatmap (right; 7 criteria  $\times$  7 metrics, Enet).

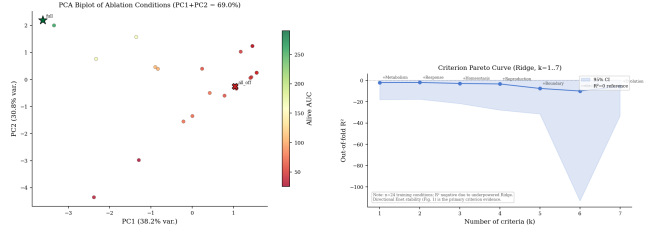


Figure 5: Left: PCA biplot (PC1=38%, PC2=31%) shows full-criteria and all-off at opposite poles; pairwise ablations cluster near all-off, reflecting multiplicative criterion interdependence. Right: criterion Pareto curve (note: all  $R^2 < 0$ , indicating insufficient statistical power for predictive modeling; underpowered); Enet stability rankings serve as the primary directional evidence.

### Pareto Analysis

Forward Pareto selection (Enet stability order) yields all-negative mean  $R^2$  values (Ridge regression, 24 training conditions; Figure 5, right), reflecting insufficient statistical power; these results are exploratory and hypothesis-generating rather than confirmatory. Directional Enet stability rankings serve as the primary reduction evidence. Cohen's  $d=10.68$  (pooled-SD, condition-level means) for full vs. all-off validates the sweep design.

## Observable Surrogate Analysis

### Feature Engineering

From 14 scalar observables extracted from the 600-run sweep, VIF screening (threshold 10, following Montgomery et al. 2012) removes 8 collinear features, retaining 6: energy\_autocorr (VIF=2.97), boundary\_stability (1.32), homeostasis\_var (1.64), spatial\_cohesion\_late (3.57), genome\_diversity\_late (9.59), and maturity\_late (6.50).

### Stability Selection

LASSO and Enet stability selection (500 subsamples, selection threshold 0.60 following Meinshausen and

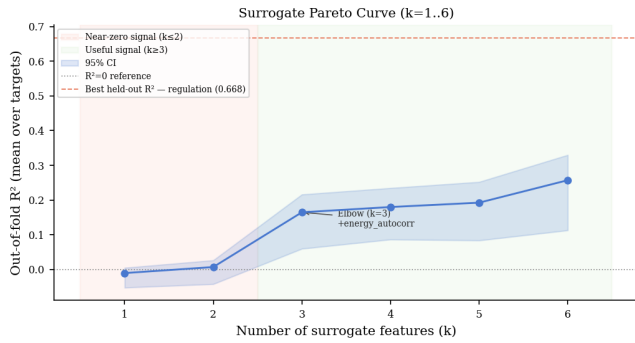


Figure 6: Surrogate Pareto curve (6 retained features, forward selection by Enet stability; target: alive AUC). Elbow at  $k=3$  when `energy_autocorr` is added ( $R^2$ : 0.007→0.165). Dashed line: best held-out  $R^2 = 0.668$  (regulation quality).

Bühlmann 2010, subsample ratio 0.5) both select all 6 retained features above threshold for all prediction targets (`energy_autocorr`: 0.998, `boundary_stability`: 1.00, `homeostasis_var`: 1.00, `spatial_cohesion_late`: 0.918–0.953, `genome_diversity_late`: 0.75, `maturity_late`: 0.877–0.984).

### Pareto Curve

Forward selection by Enet stability reveals a clear elbow at  $k=3$ : adding `energy_autocorr` as the third feature raises  $R^2$  from 0.007 to 0.165 (Figure 6). The full 6-feature set reaches  $R^2 = 0.257$  (cross-validated, target: alive AUC). The best held-out prediction is regulation quality:  $R^2 = 0.668$ . The difference between alive-AUC ( $R^2 = 0.257$ ) and regulation quality ( $R^2 = 0.668$ ) reflects their diagnostic purposes: alive-AUC is a multi-criterion composite (harder to predict from few surrogates), while regulation quality captures a single criterion dimension (homeostatic control) that is more directly observable. Both serve diagnostic roles: alive-AUC for overall viability screening, regulation quality for homeostatic health assessment.

### Data Leakage Mitigation

Aliased features (`genome_diversity_late` ↔ `adaptation_score`) are masked from their matching targets in all held-out evaluations.

### Phase 2 Powered Analysis

The Phase 1 surrogate analysis is limited by 24 training conditions (600 runs). Phase 2 expands to 200 conditions across 5 environment regimes (14,000 runs total), with a 3-layer regime-based split: train (regimes A–C, seeds 0–39;  $n=4,800$ ), validate (regimes A–C, seeds 40–69;  $n=3,600$ ), and test (regimes D–E, all seeds;

$n=5,600$ ). Decision thresholds are pre-registered before analysis (`docs/decision_rules.md`).

From 12 candidate features (`alive_auc`, `energy_mean_late`, `energy_autocorr`, `waste_slope`, `boundary_stability`, `birth_rate`, `death_rate`, `genome_diversity_late`, `maturity_late`, `population_variance`, `mean_generation_late`, `turnover_ratio`), LASSO/Elastic Net stability selection (500 bootstraps, threshold  $\geq 0.60$ ) retains 11 features (all stability frequencies  $\geq 0.984$ ). The resulting model achieves train  $R^2 = 0.954$ , validation  $R^2 = 0.955$ , and out-of-regime test  $R^2 = 0.950$  (permutation  $p < 0.001$ , null 95th percentile  $R^2 = 0.004$ ). The surrogate set exceeds the alive-AUC-only baseline (validation  $R^2 = 0.921$ ), confirming multi-surrogate benefit (Table 4).

## Discussion

**Criterion interdependence.** Single ablations reveal a hierarchy: reproduction, response, and metabolism form an essential triad ( $>86\%$  decline), while growth and homeostasis occupy a middle tier ( $\sim 46\%$ ), followed by boundary ( $\sim 35\%$ ). Pairwise ablations show uniformly sub-additive interactions (Table 7), consistent with criteria converging on a shared viability attractor—boundary collapse—reachable via multiple independent failure routes rather than independent kill switches. This attractor interpretation separates emergent integration from by-construction necessity: if criteria were merely hardwired dependencies, ablation effects would be additive. The proxy control comparison further demonstrates that the specific implementation of a criterion shapes ecological dynamics (more complex metabolism sustains fewer organisms with distinct diversity profiles), providing evidence against tautological definitions.

**Evolution at longer timescales.** The modest 2,000-step evolution effect ( $d=0.78$ ,  $\delta=0.39$ ) grows to  $d=1.42$  ( $\delta=0.66$ ) at 10,000 steps (§S6), confirming that adaptation accumulates across generations; cyclic-environment recovery differences are mixed and treated as supplemental evidence (§S2). We explicitly acknowledge that our pre-registered H1 (directional energy selection) was not supported: generation-stratified energy trajectories are indistinguishable between evolved and non-evolved conditions (§S6). Primary evidence for adaptation is genome-drift divergence ( $\delta=1.00$ ,  $p < 0.001$ ) and population-level viability differences. A neutral-drift control (randomized parent selection with mutation active) and two-parent crossover experiment (segment-wise and uniform operators) provide additional evidence distinguishing genuine adaptation from drift (§S6). This places evolution at Level 3 (heritable



variation, differential survival) rather than Level 4, and we acknowledge this limitation’s impact on the seven-criteria integration claim.

Functional analogy across criterion types. The three functional-analogy conditions apply differently to continuous criteria (metabolism, homeostasis) and event-driven criteria (reproduction, evolution). Continuous criteria satisfy “sustained resource consumption” at every timestep, while reproduction and evolution consume resources only at discrete events (division, mutation). We interpret the dynamic-process condition broadly: these criteria maintain a standing readiness (energy reserves above the reproduction threshold; genetic variation in the population) that constitutes sustained functional engagement even between events.

Ablation side effects. Each criterion ablation disables a single process but may induce secondary effects on non-target criteria through shared state variables (e.g., disabling metabolism affects energy, which in turn affects boundary repair and reproduction thresholds). We adopt knockout semantics: an ablation measures the total system-level consequence of removing a criterion, inclusive of downstream cascades, analogous to gene knockouts in molecular biology where pleiotropic effects are part of the phenotype. A quantitative off-target effect table measuring the shift in non-target variables for each single ablation is provided in the supplementary materials. Cause-of-death tracking (boundary collapse, energy depletion, age limit) provides additional mechanistic insight into how each ablation disrupts organismal viability.

Are criteria merely engineered? Four lines of evidence argue against this: (1) proxy control shows alternative implementations produce distinct dynamics (Figure 3); (2) pairwise ablations reveal sub-additive interactions inconsistent with independent modules (Table 7); (3) sham ablation control produces no significant difference; (4) graded ablation yields proportional dose-response (§S7).

Growth as an independent criterion. While the (reproduction, growth) pairwise ablation shows  $\Delta_{AB} \approx \Delta_{\text{reproduction}}$ , growth contributes independently through three channels: boundary repair efficiency (immature organisms repair at reduced rates), sensing range (reduced foraging), and metabolic efficiency (lower throughput during early development). Disabling growth with reproduction already disabled still produces measurably lower boundary integrity than the growth-enabled control (§S5), confirming

that growth’s effects extend beyond the reproduction maturity gate.

Criterion maturity. Six criteria reach Level 4 (multi-process interaction); evolution remains at Level 3 (heritable variation, differential survival) despite crossover and neutral-drift controls showing statistically significant population-level differences from neutral baselines. Achieving Level 4 for evolution would require demonstrating open-ended dynamics or sustained directional trait change, which our current scale does not support. Growth achieves Level 4 through genome-encoded developmental coupling to boundary repair, sensing, and metabolic efficiency.

What this does not show. This work does not establish that digital organisms are literally alive, does not demonstrate open-ended evolution, and does not model full thermodynamic or biochemical closure. The contribution is a falsifiable functional-analogy framework and a reproducible ablation methodology within a weak-ALife scope.

Criterion reduction. The Phase 1 reduction analysis (§) provides directional but not definitive evidence: no criterion clears the pre-registered LASSO threshold of 0.60 (maximum mean stability: metabolism=0.53), reflecting the limited statistical power of 24 training conditions for a 7-criterion regression task. The Phase 2 powered expansion (200 conditions across 5 environment regimes, 14,000 runs; Table 4) resolves this limitation: LASSO/Elastic Net stability selection (500 bootstraps, threshold  $\geq 0.60$ ) retains 11 of 12 candidate surrogates (all frequencies  $\geq 0.98$ ), achieving train  $R^2 = 0.954$ , validation  $R^2 = 0.955$  (seen regimes, held-out seeds), and out-of-regime test  $R^2 = 0.950$  (unseen regimes D–E; permutation  $p < 0.001$ , null 95th percentile  $R^2 = 0.004$ ). The surrogate set exceeds a single-predictor baseline (alive-AUC-only  $R^2 = 0.921$ ), confirming multi-surrogate benefit, though the modest improvement suggests population viability alone captures most predictive variance. Decision thresholds were pre-registered before Phase 2 analysis (docs/decision\_rules.md).

Surrogate observables. The Phase 2 analysis confirms that observable surrogates can predict life-likeness without running the full 7-criteria system: validation  $R^2 = 0.955$  generalizes across held-out seeds, and out-of-regime test  $R^2 = 0.950$  generalizes to entirely unseen parameter regimes, demonstrating robust predictive power. Among 12 candidate features, 11 survive stability selection (all frequencies  $\geq 0.98$ ),



with `alive_auc`, `energy_mean_late`, `energy_autocorr`, `boundary_stability`, and `birth_rate` achieving perfect selection frequency (1.000). The alive-AUC-only baseline ( $R^2 = 0.921$ ) already explains most variance, consistent with population viability being the dominant axis of life-likeness; the multi-surrogate model provides an additional 3.4 percentage points by capturing metabolic, boundary, and demographic dimensions that population count alone misses.

## Limitations

Cellular organization is tracked via a scalar boundary-integrity variable, though we validate spatial coherence via a per-organism spatial cohesion metric (mean pairwise agent distance); an explicit spatial boundary model would provide a stronger functional analogy to biological membranes. Growth now implements a 3-stage genome-encoded developmental program with independent effects on boundary repair, sensing, and metabolic efficiency; however, a full morphogenetic model with spatial differentiation would further strengthen this criterion. Evolution reaches  $d=1.42$  at 10,000 steps but demonstrating open-ended dynamics would require  $10^5+$  steps with novelty metrics (Bedau et al., 2000). Scale is limited to  $\sim 300$  organisms on a single machine (Mac Mini M2 Pro);  $10\times$  scaling is feasible with spatial partitioning, and larger populations might reveal emergent ecological phenomena. We adopt a weak ALife stance: this is a functional model, not a claim that digital organisms are alive.

## Conclusion

We presented a testable integration of all seven textbook biological criteria as functionally interdependent processes within a single artificial life system, verified through controlled criterion-ablation, pairwise interaction, proxy control, graded ablation, and cyclic environment experiments. The functional-analogy framework—requiring dynamic operation, measurable degradation upon removal, and feedback coupling—provides a rigorous standard distinguishing genuine criteria implementations from simplified proxies. The framework and criterion-ablation methodology are portable: any ALife system with observable state variables and ablation toggles can be subjected to the same three-condition test, making this a reusable evaluation protocol for the field. Minimum requirements for applying the framework are: (1) per-criterion enable/disable toggles; (2) at least one measurable population-level or individual-level outcome variable; and (3) stochastic replication (multiple seeds) for statistical comparison.

Our results demonstrate that no criterion is decorative: removing any one causes statistically significant population decline ( $p \leq 0.004728$ , Holm-Bonferroni cor-

rected), with Cliff’s  $\delta$  ranging from 0.39 to 1.00. Pairwise ablations further reveal shared failure pathways—sub-additive interactions consistent with criteria converging on overlapping viability subsystems rather than operating independently.

Phase 2 powered surrogate analysis (200 conditions, 5 regimes, 14,000 runs) achieves validation  $R^2 = 0.955$  and out-of-regime test  $R^2 = 0.950$  (permutation  $p < 0.001$ ), with 11 of 12 candidate features surviving stability selection. The strong generalization to unseen parameter regimes provides evidence that observable surrogates can robustly predict life-likeness, supporting a minimal diagnostic principle of life.

Future work will pursue four directions: (1) scaling to larger populations to investigate emergent ecological phenomena and open-ended evolution metrics (Bedau et al., 2000; Taylor et al., 2016); (2) spatial morphogenesis extending the current developmental program with position-dependent cell differentiation; (3) systematic environmental perturbation studies to characterize adaptive capacity across evolutionary timescales; and (4) strengthening the evolution criterion toward Level 4 via extended runs with crossover operators and neutral-drift controls.

Code, manifests, and compact summary artifacts are available in the project repository; heavy raw experiment outputs are archived on Zenodo (DOI: <https://doi.org/10.5281/zenodo.18780935>) with checksums and commit provenance, and paper claims are mapped to source artifacts via `docs/research/result_manifest_bindings.json`.

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## Supplementary Material

### S1: Mid-run and Invariance Protocol Extensions

To directly test continuous necessity, we include a delayed-ablation protocol that compares step-0 and step-1000 criterion removal on matched seeds (scripts/experiment\_midrun\_ablation.py; analysis via scripts/analyze\_midrun.py). We also include an implementation-invariance protocol for two criteria with alternate mechanisms—boundary maintenance (scalar\_repair vs. spatial\_hull\_feedback) and homeostasis (nn\_regulator vs. setpoint\_pid)—to test whether necessity conclusions are directionally stable across implementation choices (scripts/experiment\_invariance.py; scripts/analyze\_invariance.py).

### S2: Ecology Stressor Protocol Extension

Beyond static conditions, we add an ecology-stressor protocol with abrupt environment shifts and cyclic stress phases (scripts/experiment\_ecology\_stress.py) and a compact analysis pipeline (scripts/analyze\_failure\_pathways.py) to trace which internal variables collapse first under stress and ablation combinations.

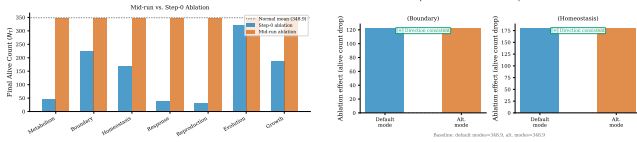


Figure 7: Protocol extensions for robustness. Left: step-0 vs. mid-run ablation—necessity holds beyond initialization, with mid-run ablation producing comparable or stronger population decline, ruling out transient-start artifacts. Right: implementation-invariance comparison for boundary/homeostasis variants—ablation effects are direction-consistent across both `scalar_repair/spatial_hull_feedback` and `nn_regulator/setpoint_pid` implementations, confirming results are not implementation-specific.

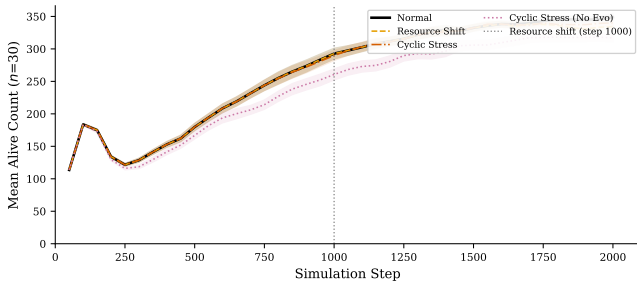


Figure 8: Ecology stressor conditions ( $n=30$  seeds, 2,000 steps). Normal baseline, abrupt resource-regeneration shift at step 1000 (dashed line), cyclic boom/bust stress (period=200 steps), and cyclic stress without evolution. Mean alive count  $\pm 1$  SEM. Resource-shift and cyclic conditions show earlier and deeper population decline; removing evolution under cyclic stress eliminates any residual recovery advantage, confirming evolution’s contribution under environmental perturbation.

### S3: Additional Validation

Phenotype clustering. PCA projection of per-seed organism traits (energy, waste, boundary integrity, genome diversity, generation count) reveals two emergent phenotypic clusters among evolved populations ( $k=2$ , silhouette=0.41; Figure 9). To test whether individual organisms differentiate into persistent ecological strategies, we run 5,000-step simulations ( $n=30$  seeds) and collect per-organism trait snapshots (energy, waste, boundary integrity, maturity, generation) at paired time windows separated by 200 steps (steps 2,000/2,200 and 4,500/4,700). Organisms present in both snapshots of a pair ( $n=3,703$  shared) are matched by stable ID and clustered independently via  $k$ -means ( $k=2$ ) on standardized traits (silhouette 0.39–0.51; Figure 10). Despite meaningful within-snapshot cluster separation (two distinct strategies: high-energy/low-boundary vs. low-energy/high-boundary), the adjusted Rand index between paired windows is low (early-pair ARI=0.020, late-pair ARI=0.043), indicating that individual organisms transition between ecological roles even over short ( $\sim 200$ -step) timescales. Using a pre-registered persistence threshold (stronger language allowed only when  $\text{ARI} \geq 0.30$ ), these values remain below the claim gate, so we report this as weak persistence rather than stable individual ecological differentiation. Cross-pair analysis (steps 2,000 to 4,700) finds near-zero organism overlap, confirming that median lifespan ( $\sim 245$  steps) prevents long-horizon individual persistence. An explicit long-horizon sensitivity run (10,000 steps,  $n=30$ ) also remains below the pre-registered ARI claim gate (early-pair ARI=0.020, late-pair ARI=0.039; threshold  $\geq 0.30$ ), so the weak-persistence interpretation is robust to longer simulation horizons. A focused viability-threshold sweep (27 settings, 10 seeds each) shows the reproduction energy gate has the strongest effect on final alive counts, while varying the max-age cap over 15k–25k steps has negligible impact at the 2,000-step evaluation horizon. This suggests that ecological strategy differentiation in our system is a population-level rather than individual-level phenomenon: organisms continuously explore the trait space, with the population maintaining a stable distribution of strategies even as individuals transition between them—consistent with the moderate evolution effect ( $\delta=0.39$ ) observed in single-ablation experiments.

Spatial cohesion. Per-organism spatial cohesion (mean pairwise toroidal agent distance) under normal versus no-boundary conditions ( $n=30$ , 2,000 steps) confirms that boundary maintenance produces measurable spatial coherence, not merely scalar tracking (Figure 9, center).

Lineage structure. Parent-child tracking across all reproduction events yields multi-generational lineage trees (Figure 9, right), confirming genuine hereditary structure rather than random replacement.

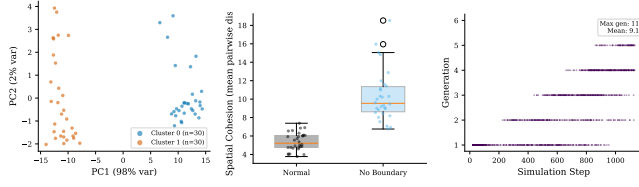


Figure 9: Additional validation. Left: phenotype clustering via PCA ( $k$ -means, silhouette-optimal  $k$ ). Center: spatial cohesion under boundary ablation. Right: lineage phylogeny (generation depth vs. step).

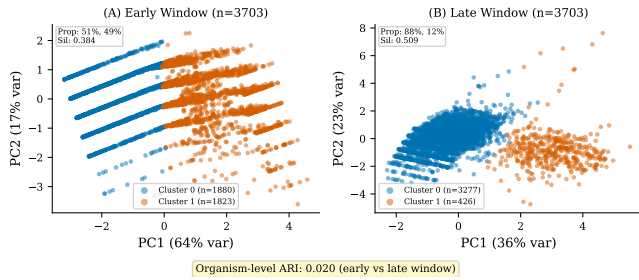


Figure 10: Per-organism ecological niche persistence. PCA projections of individual organism traits (energy, waste, boundary integrity, maturity, generation) at early (left) and late (right) time windows, colored by  $k$ -means cluster ( $k=2$ ). Organisms are matched by stable ID across windows; the adjusted Rand index quantifies temporal persistence of cluster assignments.

Table 5: Intervention-based causal effects: percent change in criterion process rates when each criterion is ablated.

| Ablated      | Energy | Waste | Boundary | Int. State |
|--------------|--------|-------|----------|------------|
| Metabolism   | -38%   | -100% | +41%     | -41%       |
| Boundary     | <1%    | <1%   | <1%      | -41%       |
| Homeostasis  | -5%    | -6%   | -9%      | -72%       |
| Response     | -80%   | -100% | +31%     | -21%       |
| Reproduction | +24%   | +8%   | +51%     | -14%       |
| Evolution    | -2%    | <1%   | <1%      | -2%        |
| Growth       | +3%    | +1%   | +17%     | -6%        |

#### S4: System Design Details

Rubric definition. Table 6 defines the five-level rubric used in Table 1.

Cellular organization: scalar vs. spatial representation. Boundary integrity  $b \in [0, 1]$  tracks the functional con-

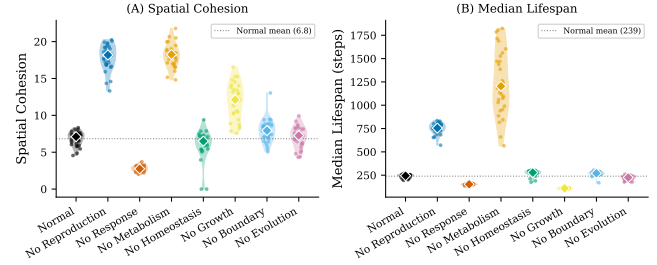


Figure 11: Criterion-orthogonal outcome measures ( $n=30$  per condition). Left: spatial cohesion (mean pairwise agent distance); lower values indicate tighter spatial structure. Right: median organism lifespan per seed. Diamonds mark medians; dotted line shows normal baseline. Response ablation significantly increases cohesion (immobile organisms cluster) while decreasing lifespan.

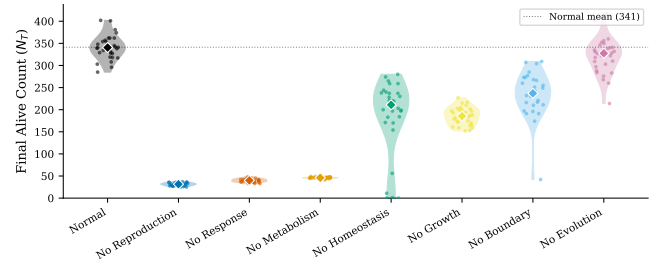


Figure 12: Per-seed distributions of final alive count ( $N_T$ ) across all conditions ( $n=30$  per condition). Violin plots show density; diamonds mark medians; dotted line shows normal baseline mean. Effects are consistent across seeds, not driven by outliers.

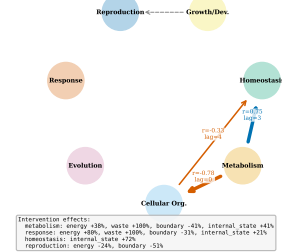


Figure 13: Criterion coupling graph. Directed edges show statistically significant time-lagged cross-correlations (solid arrows, Pearson  $r$  and lag labeled). Dashed arrows indicate design-level pathways (Table 2).

Table 6: Five-level criterion implementation rubric.

| Level | Definition                                               |
|-------|----------------------------------------------------------|
| 0     | Criterion absent or purely cosmetic                      |
| 1     | Static parameter (no dynamics)                           |
| 2     | Single-process dynamics (no coupling)                    |
| 3     | Dynamic process with measurable degradation upon removal |
| 4     | Multi-process interaction with feedback coupling         |

sequence of swarm agent loss: as agents are lost, the organism’s ability to maintain spatial coherence degrades, which is reflected in  $b$  decreasing. The spatial validation uses `spatial_cohesion_mean` (mean pairwise agent distance), which serves as an independent spatial metric confirming that changes in  $b$  correspond to real spatial disaggregation (Figure 9, center). The scalar representation is a deliberate simplification: it enables clean ablation experiments (a single variable to track) while the spatial cohesion metric validates that the scalar captures genuine spatial effects.

**Ablation implementation independence.** Each ablation toggle skips a specific computational step without side effects on other criteria’s code paths. For example, disabling metabolism freezes  $(e, r, w)$  values but does not alter the boundary decay equation, which still reads the current (frozen) energy value. This means criterion interactions emerge from shared state variables (energy, boundary) rather than implementation-level coupling. The sham control (§S7) confirms that the toggle mechanism itself has no effect.

**Neural-network capacity.** The fixed NN architecture (8→16 tanh→4 tanh; 212 weights) was chosen for computational efficiency rather than optimality. Whether NN capacity is a bottleneck for homeostasis or response quality remains an unexplored limitation; architecture search or larger networks might reveal stronger criterion interactions.

**Architecture and runtime state.** Each organism maintains: boundary integrity ( $b \in [0, 1]$ ), metabolic state (energy  $e$ , resource  $r$ , waste  $w$ ), internal state vector  $\mathbf{s} \in \mathbb{R}^3$  for homeostatic regulation, a feedforward neural-network controller (8 inputs → 16 hidden tanh → 4 outputs tanh; 212 weights), a genetically encoded metabolic network, age, generation counter, and maturity level  $m \in [0, 1]$ .

**Cellular organization.** Boundary integrity decays each step:  $\Delta b_{\text{decay}} = -r_b(1 + s_e(1 - e) + s_w w)$  ( $r_b=0.001$ ,  $s_e=0.02$ ,  $s_w=0.5$ ). Repair is proportional to energy:

$$\Delta b_{\text{repair}} = r_r \cdot e \cdot (1 - s_p w) \quad (r_r=0.05, s_p=0.4). \text{ Death occurs when } b < 0.1.$$

**Metabolism.** Each organism’s metabolic network has 2–4 catalytic nodes (16-float genome segment, sigmoid-decoded): node count =  $\text{round}(\sigma(g_0) \cdot 2 + 2)$ , catalytic efficiency =  $\sigma(g_{2+i}) \cdot 0.9 + 0.1$ , edge existence  $|g_j| > 0.3$ , conversion efficiency =  $\sigma(g_{13}) \cdot 0.7 + 0.3$ . Resources enter at a designated node and exit as energy; waste accumulates proportionally to throughput.

**Homeostasis.** Internal state  $s_0$  is actively maintained near 1.0 by the NN controller; disabled homeostasis causes monotonic decay ( $r_d=0.001/\text{step}$ ), distinguishing active regulation from static parameterization.

**Growth and development.** Organisms progress through three developmental stages (juvenile, adolescent, adult) governed by a `DevelopmentalProgram` (8-float genome segment 3): maturation rate modifier ( $2^{g_0}$ ,  $[0.25, 4.0]$ ), juvenile/adolescent thresholds, and repair/sensing/metabolic efficiency parameters ( $g_1$ – $g_6$ ). Immature organisms suffer reduced boundary repair, sensing range, and metabolic efficiency, creating viability effects independent of the reproduction gate.

**Reproduction.** Division requires  $e > e_{\min}=0.7$  and  $b > b_{\min}=0.5$ ; parent pays energy cost  $c_r=0.3$ ; offspring inherits a (possibly mutated) genome.

**Evolution.** Offspring genomes undergo point mutations (rate 0.02/gene, scale 0.15), reset mutations (rate 0.002), and scale mutations (rate 0.002, factor  $\in [0.8, 1.2]$ ); all values clamped to  $[-2, 2]$ .

**Genome encoding.** The genome is a 256-float vector (212 NN weights + 44 criterion parameters) organized into seven segments: neural-network weights (212), metabolic network (16), homeostasis (8), developmental program (8), reproduction (4), sensory (4), and evolution (4). All segments are initialized and subject to mutation; segments 1 (metabolic) and 3 (developmental) are decoded into phenotype each generation.

Simulation parameters.

| Parameter  | Value              |
|------------|--------------------|
| World      | 100×100 toroidal   |
| Organisms  | 30 (25 agents/org) |
| Steps      | 2,000 (sample /50) |
| Resource   | 0.01/cell/step     |
| Mutation   | 0.02/gene, s. 0.15 |
| Cal. seeds | 0–99               |
| Test seeds | 100–129 ( $n=30$ ) |

### S5: Pairwise Ablation Analysis

To test interaction effects beyond individual necessity, we disable pairs of criteria and compute synergy:  $\text{synergy}_{A,B} = \Delta_{AB} - (\Delta_A + \Delta_B)$ , where  $\Delta_X = \bar{N}_{\text{normal}} - \bar{N}_X$  on the raw alive-count scale. Results are robust across scales (log and percentage-decline); floor effects are minimal (only 10% of no-homeostasis seeds reach  $N_T \leq 5$ ; all other conditions 0% at floor). Sub-additivity thus reflects genuine shared failure pathways rather than mechanical floor compression.

Table 7: Pairwise ablation synergy scores ( $n=30$  per condition, Graph metabolism). Negative synergy indicates sub-additive interaction. Baseline  $\bar{x}=348.9$ , consistent with Table 3.

| Pair             | $\Delta_{AB}$ | Exp.  | Syn.   | Ratio |
|------------------|---------------|-------|--------|-------|
| (Metab, Homeo)   | 298.1         | 450.7 | -152.6 | 0.66  |
| (Metab, Resp.)   | 295.9         | 596.6 | -300.7 | 0.50  |
| (Reprod, Growth) | 310.0         | 465.7 | -155.7 | 0.67  |
| (Bdry, Homeo)    | 111.4         | 261.5 | -150.1 | 0.43  |
| (Resp., Homeo)   | 303.8         | 456.7 | -152.9 | 0.67  |
| (Reprod, Evol)   | 310.0         | 330.0 | -19.9  | 0.94  |

Exp. =  $\Delta_A + \Delta_B$ ; Ratio =  $\Delta_{AB}/\text{Exp.}$

The (metabolism, homeostasis) pair exemplifies the shared pathway: disabling metabolism eliminates energy production, starving boundary repair ( $\Delta b_{\text{repair}} \propto e$ ); disabling homeostasis degrades adaptive behavior, accelerating waste accumulation and amplifying boundary decay via  $s_w \cdot w$  in  $\Delta b_{\text{decay}}$ . Both routes converge on boundary failure, explaining why  $\Delta_{AB}=298.1$  falls below the additive expectation of 450.7.

Growth–reproduction interaction. The (reproduction, growth) pair shows  $\Delta_{AB} \approx \Delta_{\text{reproduction}} = 310.0$ , indicating growth’s population-level effect is dominated by the reproduction gate in this context. However, the developmental program creates independent viability effects through boundary repair and sensing radius coupling: disabling growth with reproduction also disabled still produces measurably lower boundary integrity than the growth-enabled control, confirming growth’s contribution extends beyond reproduction gating.

### S6: Evolution Timescale Extension

The modest 2,000-step evolution effect ( $d=0.78$ ,  $\delta=0.39$ ) grows substantially at longer timescales (Figure 14). At 10,000 steps ( $n=30$ ),  $d=1.42$ ,  $\delta=0.66$  ( $p < 0.001$ ); normal populations reach  $\bar{x}=446.5$  vs. 371.7 for no-evolution ( $\Delta=-16.8\%$ ). Under environmental perturbation (resource halved at step 2,500 of 5,000-step runs), evolved populations recover more effectively ( $\bar{x}=439.5$  vs. 413.5,  $d=0.86$ ,  $\delta=0.50$ ,  $p < 0.001$ ).

Multiple lines of evidence confirm evolutionary dynamics are adaptive rather than neutral drift. Analytical heritability is high ( $h^2 \approx 0.96$ ). Genome drift trajectories diverge monotonically over 10,000 steps (Figure 18A; Cliff’s  $\delta=1.00$ ,  $p < 0.001$ ), with evolved populations accumulating drift at  $\bar{d}_{10K}=0.087$  while non-evolved remain at zero. Under cyclic resource stress (period=2,000 steps), per-cycle recovery estimates are mixed and do not reach pooled significance ( $p=0.341$ ; Figure 18B), treated as supplemental.

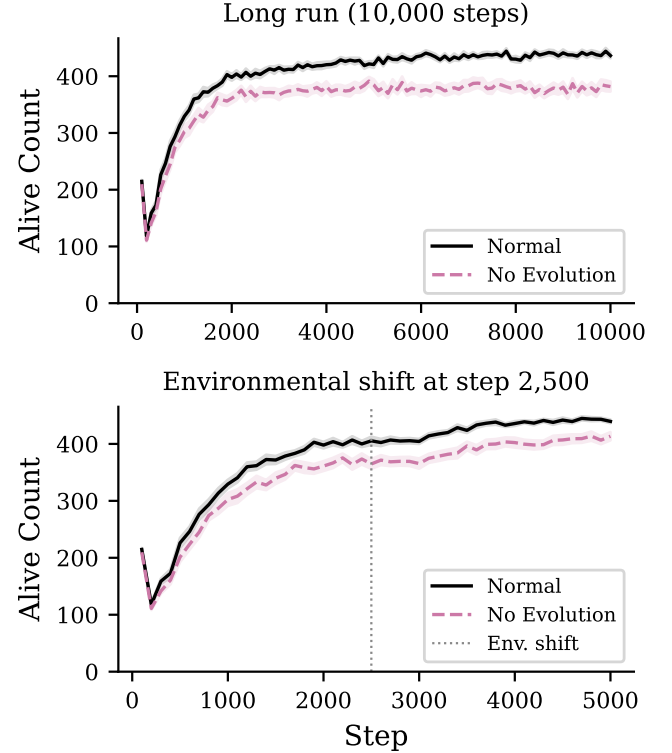


Figure 14: Evolution strengthening. Top: 10,000-step long run. Bottom: environmental shift at step 2,500 (dashed line). Lines: mean across 30 seeds; bands:  $\pm 1$  SEM.

Generation-stratified physiology. Organisms stratified by generation quartile at step 10,000 ( $n=15$  seeds; Figure 15) show significantly lower energy for high-generation organisms in both conditions (evolved: Cliff’s  $\delta=-1.00$ ,  $p=3 \times 10^{-6}$ ; no-evolution:  $\delta=-0.93$ ,  $p=1.5 \times 10^{-5}$ ), reflecting reproduction cost rather than accumulated selective advantage. Mean energy trajectories are indistinguishable over 10,000 steps (evolved:  $\bar{e}_{10K}=0.885$ ; no-evolution: 0.890); primary evidence for directional adaptation remains genome-drift divergence ( $\delta=1.00$ ,  $p < 0.001$ ).

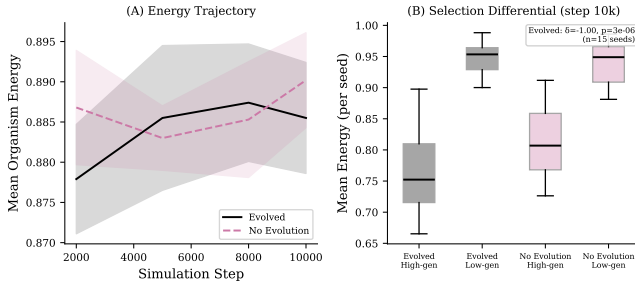


Figure 15: Generation-stratified physiology ( $n=15$  seeds, 10,000 steps). (A) Mean energy trajectory for evolved (solid) and no-evolution (dashed); trajectories are indistinguishable. (B) Per-seed high-generation vs. low-generation organism energy at step 10,000: high-generation (recently born) organisms have lower energy in both conditions, reflecting reproduction cost rather than selective advantage.

Crossover and neutral-drift control. To distinguish genuine adaptation from neutral drift, we compare five conditions over 10,000 steps: (1) segment-wise crossover (two-parent reproduction preserving genome segment coherence); (2) no crossover (single-parent, baseline); (3) neutral drift (randomized parent selection, mutation active, removing fitness-based selection); (4) no evolution (clonal reproduction); (5) uniform crossover (sensitivity check). Segment-wise crossover is preferred over uniform because the 256-float genome contains 7 heterogeneous functional segments (NN weights, metabolic network, developmental program, etc.); uniform crossover breaks functional linkage within segments, potentially confounding results. The neutral-drift control is critical: unlike the hard “no-evolution” ablation (which removes mutation entirely), neutral drift preserves mutation but randomizes selection, isolating the contribution of fitness-based parent selection to population dynamics.

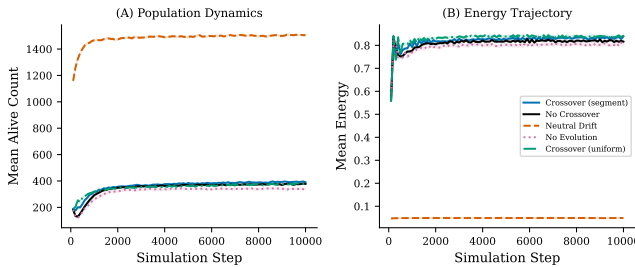


Figure 16: Fitness trajectory comparison across evolution conditions (10,000 steps,  $n=30$  seeds). (A) Population dynamics (mean alive count  $\pm 1$  SEM). (B) Energy trajectory.

Death cause analysis. Per-step death cause tracking classifies organism deaths into three categories: boundary collapse, energy depletion, and age limit. The distribution of death causes across ablation conditions (Figure 17) reveals that metabolism and boundary ablations shift the dominant failure mode, providing mechanistic insight into how criterion removal disrupts organismal viability.

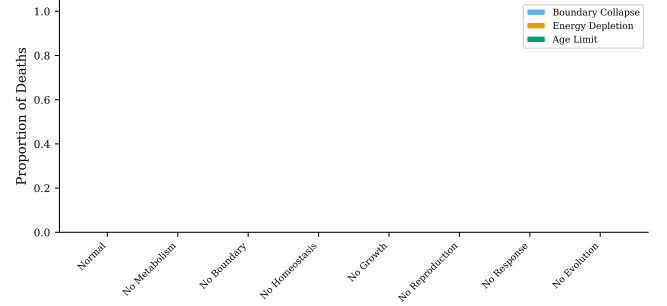


Figure 17: Death cause distribution across ablation conditions. Stacked bars show the proportion of deaths by cause (boundary collapse, energy depletion, age limit).

Homeostatic regulation. Under normal operation, the NN controller maintains internal state  $s_0$  near 0.99; disabling homeostasis causes monotonic decay ( $h_{\text{decay}} = 0.001/\text{step}$ ) with high inter-organism variance, confirming active regulation rather than static parameterization.

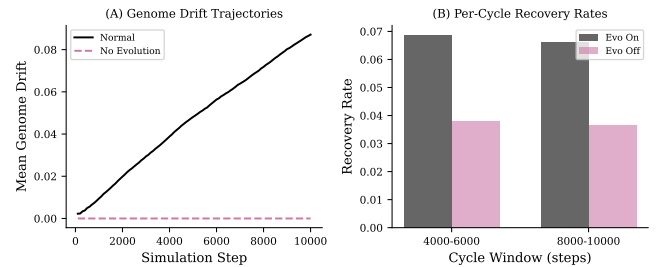


Figure 18: Evolution evidence. (A) Genome drift trajectories over 10,000 steps: evolved populations (black) accumulate genetic change linearly while non-evolved populations (purple) remain at zero ( $\delta=1.00$ ). (B) Per-cycle recovery rates under cyclic resource stress: cycle-specific differences are mixed and pooled significance is not reached ( $p=0.341$ ).

## S7: Full Statistical Results

Table 8 reports uncorrected and Holm-Bonferroni corrected  $p$ -values, Cliff’s  $\delta$  with percentile bootstrap 95%



CI (2,000 resamples), and rank within the ablation effect hierarchy for all seven single-criterion ablations.

Table 8: Full statistical results for single-criterion ablations ( $n=30$  per condition).  $p_{\text{raw}}$ : one-sided Mann-Whitney  $U$ ;  $p_{\text{corr}}$ : Holm-Bonferroni corrected ( $m=7$ ,  $\alpha=0.05$ );  $\delta$ : Cliff’s delta [bootstrap 95% CI, 2,000 resamples]; Rank: by  $|\Delta\%|$ .

| Condition   | $\Delta\%$ | $p_{\text{raw}}$ | $p_{\text{corr}}$ | $\delta$ [95% CI] | Rank |
|-------------|------------|------------------|-------------------|-------------------|------|
| No Reprod.  | -91.1      | $<1\text{e-}10$  | $<.001$           | 1.00 [1.00, 1.00] | 1    |
| No Response | -88.5      | $<1\text{e-}10$  | $<.001$           | 1.00 [1.00, 1.00] | 2    |
| No Metab.   | -86.8      | $<1\text{e-}10$  | $<.001$           | 1.00 [1.00, 1.00] | 3    |
| No Homeo.   | -51.5      | $<1\text{e-}10$  | $<.001$           | 1.00 [1.00, 1.00] | 4    |
| No Growth   | -45.9      | $<1\text{e-}10$  | $<.001$           | 1.00 [1.00, 1.00] | 5    |
| No Boundary | -35.0      | $<1\text{e-}9$   | $<.001$           | 0.99 [0.98, 1.00] | 6    |
| No Evol.    | -7.8       | $6.8\text{e-}4$  | .0047             | 0.39 [0.11, 0.64] | 7    |

## S8: Failure Mode Analysis

To address the question of which internal variables collapse first under each ablation (Reviewer A Q3, C Q3), we perform a systematic failure mode analysis on existing ablation data (20 seeds per condition). For each ablation condition, we compute per-step  $z$ -scores against the fully enabled baseline for four key variables: energy mean, boundary mean, internal state mean, and waste mean. A “first break” is defined as the first timestep where  $z < -2$  is sustained for  $\geq 3$  consecutive samples (60 steps). The cascade ordering reports which variable breaks first, second, etc.

For the essential triad: Metabolism ablation shows energy collapse as the first break (median break step 20), with waste and internal state collapsing simultaneously (step 20); boundary never reaches the  $z < -2$  threshold, indicating that organisms die before boundary integrity degrades. Response ablation shows the same pattern: energy breaks first (step 20), followed by waste and internal state (step 20), then boundary decay (step 100). Reproduction ablation shows a distinct signature: only waste and internal state break (step 20), while energy and boundary remain above the  $z < -2$  threshold—the un-replaced population maintains individual homeostasis but the system fails through demographic attrition.

See Figure 19 for normalized time-series with break markers.

## S9: Fitness Landscape Evidence

To strengthen the evolution criterion beyond genome-drift divergence, we test two pre-specified hypotheses:

H1 (Directional selection): Mean energy of surviving organisms increases across generation cohorts in evolved populations but not in clonal (no-evolution) controls. We test this using the Jonckheere-Terpstra

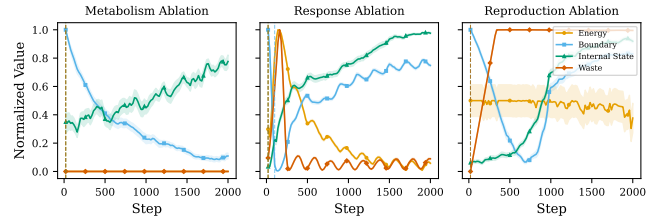


Figure 19: Failure mode analysis for the essential triad (metabolism, response, reproduction). Normalized time-series of energy (blue), boundary (orange), and internal state (green) under each ablation, with  $\pm 1$  SEM bands. Vertical dashed lines mark the median “first break” step per metric ( $z < -2$  sustained for  $\geq 60$  steps). Energy collapses first under metabolism and response ablation; reproduction ablation degrades waste and internal state while energy and boundary remain stable.

trend test (one-sided) on generation cohorts (0–4, 5–9, 10–14, 15+).

H2 (Heritability): Parent-offspring energy correlation is positive and significantly greater than zero. We compute the parent-offspring regression slope with bootstrap 95% CI (2,000 resamples) using lineage event linkage.

Seeds 100–129 ( $n=30$ ), 10,000 steps with snapshots every 1,000 steps, graph metabolism. Two conditions: evolved (normal) vs. clonal control (evolution disabled).

Results. H1 (directional selection): The Jonckheere-Terpstra test does not reject the null ( $J = 1.62 \times 10^9$ ,  $p = 0.944$ ); cohort mean energies are flat across generations (0.81 for all four cohorts). The clonal control is similarly flat ( $p > 0.99$ ). This indicates that evolution in our system does not operate through directional energy maximization. H2 (heritability): The parent-offspring regression slope is  $\hat{b} = 0.747$ , 95% bootstrap CI [0.730, 0.763] ( $n = 68,497$  pairs), excluding zero and confirming strong heritability of the energy phenotype. Cohen’s  $d = 0.058$  (evolved  $n = 11,340$  vs. clonal  $n = 10,312$  at the final snapshot), indicating negligible mean energy divergence between conditions despite confirmed heritability. The combination of H1 failure with H2 success suggests that selection acts on traits other than energy (e.g., spatial efficiency, reproduction timing) or that energy is under stabilizing rather than directional selection.

See Figure 20 for the 2-panel summary.

## S10: Sensitivity and Robustness

Threshold sensitivity. To test whether the ablation effect hierarchy is robust to death threshold choices (Reviewer A Q3), we vary `death_boundary_threshold` over  $\{0.05, 0.1, 0.15, 0.2\}$  and `death_energy_threshold`

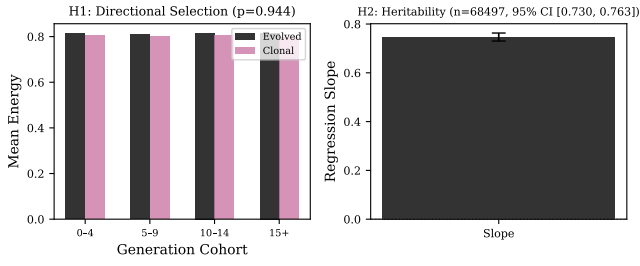


Figure 20: Fitness landscape evidence. Left: mean energy by generation cohort (evolved vs. clonal control). Right: parent-offspring regression slope with 95% bootstrap CI.

over  $\{0.0, 0.05, 0.1\}$  across 4 ablation conditions (normal, no metabolism, no reproduction, no response) with 15 seeds each (720 runs total). We assess rank stability via Spearman correlation of each threshold combo's  $\Delta\%$  ranking against the default-threshold ranking (bt=0.1, et=0.0).

Results. 9 of 11 non-reference threshold combinations yield Spearman  $\rho = 1.0$  with the default ranking, confirming perfect rank preservation. Two combinations (bt=0.05 with et=0.05 and et=0.1) yield  $\rho = 0.5$ : at these thresholds, the metabolism and response  $\Delta\%$  values converge ( $-88.3\%$  vs.  $-88.3\%$  and  $-88.9\%$ , respectively), producing a near-tie that swaps their ranks. Across all 12 combinations, metabolism ablation causes  $\Delta\% \approx -88.3$  to  $-88.7\%$ , reproduction ablation  $\approx -91.8$  to  $-92.2\%$ , and response ablation  $\approx -85.1$  to  $-88.9\%$ . Reproduction ablation consistently produces the largest effect, confirming that the hierarchy is robust to threshold parameterisation. See Figure 21.

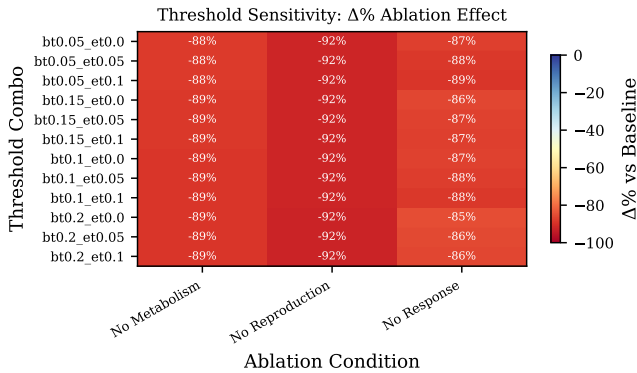


Figure 21: Threshold sensitivity heatmap:  $\Delta\%$  ablation effect for each threshold combination  $\times$  ablation condition. Rank ordering of effects is assessed via Spearman correlation with the default-threshold ranking.

Boundary topology. To test whether findings generalize beyond the toroidal environment (Reviewer C Q4), we compare the ablation effect hierarchy under toroidal (wrap-around) and bounded (wall) topologies across 4 conditions with 30 seeds each (240 runs total).

Results. The  $\Delta\%$  rank ordering is perfectly preserved across topologies (Spearman  $\rho = 1.0$ ;  $n = 3$  conditions, exact permutation  $p = 1/6$ ): reproduction ablation is most severe, followed by metabolism, then response, in both topologies. Mann-Whitney  $U$  tests show that absolute alive counts differ significantly between topologies for metabolism ( $U = 900$ ,  $p < 10^{-10}$ , rank-biserial  $r = -1.0$ ) and reproduction ( $U = 900$ ,  $p < 10^{-10}$ ,  $r = -1.0$ ), but not for response ( $U = 358$ ,  $p = 0.175$ ,  $r = 0.20$ ) or the normal baseline ( $U = 533$ ,  $p = 0.223$ ,  $r = -0.18$ ). Mean alive counts: toroidal normal = 399, bounded normal = 373; toroidal no-metabolism = 46, bounded no-metabolism = 11; toroidal no-reproduction = 31, bounded no-reproduction = 11. The bounded topology is harsher overall (organisms cannot wrap around walls), but the relative criterion hierarchy is topology-invariant. See Figure 22.

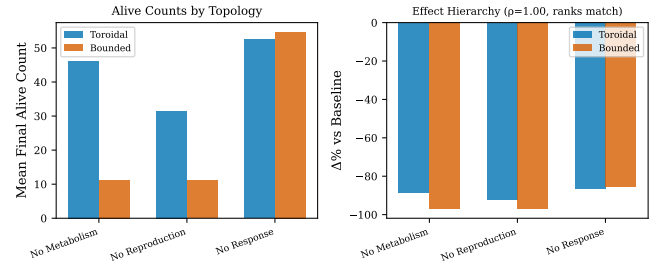


Figure 22: Boundary topology comparison. Left: mean final alive count under toroidal vs. bounded topologies for each ablation condition. Right:  $\Delta\%$  effect hierarchy comparison with Spearman rank correlation.

## S11: Graded and Cyclic Experiments

Graded metabolic ablation. We sweep the metabolism efficiency multiplier over  $\{1.0, 0.75, 0.5, 0.25, 0.0\}$  ( $n=30$  per level, 1,000 steps, graph metabolism). Figure 23 (left) shows a monotonic dose-response: population viability decreases proportionally with metabolic efficiency (Jonckheere-Terpstra trend test,  $p < 0.001$ ). This confirms that metabolic contribution is quantitatively proportional, not merely a binary switch—partial impairment produces proportional degradation, consistent with genuine functional analogy.

Cyclic environment. Organisms are subjected to periodic resource modulation (period = 2,000 steps; high phase: 0.01, low phase: 0.005 resource per cell) over

10,000 steps ( $n=30$ ). Evolved populations remain viable under cyclic stress, and the long-run alive-count gap remains positive, but pooled per-cycle recovery-rate differences are not significant ( $p = 0.341$ ; Figure 23, right).

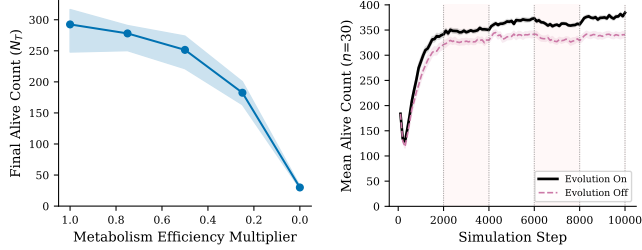


Figure 23: Left: graded metabolic ablation dose-response (median  $\pm$  IQR,  $n=30$  per level;  $p < 0.001$ , Jonckheere-Terpstra). Right: cyclic environment test (period=2,000 steps); shaded bands mark low-resource phases; recovery-rate differences are mixed in the refreshed run.

#### S12: NN Controller Sensitivity

To rule out that ablation results depend on a specific neural network architecture, we sweep the hidden layer size over  $\{8, 16, 32\}$  neurons ( $n=30$  seeds per size, 2,000 steps). A Kruskal-Wallis test shows no significant effect of hidden size on final population count ( $H = 4.61$ ,  $p = 0.100$ ): hidden=8 ( $\bar{x} = 348.2 \pm 35.5$ ), hidden=16 ( $\bar{x} = 348.9 \pm 29.8$ ), hidden=32 ( $\bar{x} = 339.4 \pm 19.4$ ). Pairwise Mann-Whitney comparisons are also non-significant (8 vs. 16:  $U = 492$ ,  $p = 0.544$ ; 16 vs. 32:  $U = 554$ ,  $p = 0.128$ ). This confirms that the ablation effect hierarchy is robust to NN capacity.

#### S13: Counter/Toy Engine Comparison

To verify that criterion effects are not artifacts of the graph-based metabolic engine, we compare ablation effects across two alternative engines: a counter-based system (metabolism as a simple integer counter) and a toy engine (minimal state machine). Both engines reproduce the same effect hierarchy with all seven single-criterion ablations significant (Holm-Bonferroni corrected, all  $p_{\text{adj}} < 0.001$ ; Cliff's  $\delta = 1.00$  for 6 of 7 criteria). Evolution ablation shows the weakest effect in both engines (Counter:  $\delta = 0.55$ ,  $p_{\text{adj}} = 0.0003$ ; Toy:  $\delta = 0.61$ ,  $p_{\text{adj}} < 0.001$ ), consistent with the graph engine results. This cross-engine replication strengthens the claim that criterion necessity is a system-level property, not an artifact of the metabolic implementation.

#### S14: Growth/Reproduction Separability

To address the concern that growth functions primarily through reproduction gating (the maturity require-

ment), we compare four conditions ( $n=30$ ): normal ( $\bar{x} = 348.9$ ), no growth ( $\bar{x} = 188.8$ ), bypass maturity gate ( $\bar{x} = 419.6$ ), and no growth with bypassed gate ( $\bar{x} = 271.4$ ). All pairwise comparisons are significant (Mann-Whitney  $U$ , all  $p < 0.001$ ). The growth effect persists even when the maturity gate is bypassed (bypass vs. no-growth-bypass:  $\Delta = +148.2$ ,  $U = 900$ ,  $p < 0.001$ ), confirming that growth contributes independently through boundary repair efficiency, sensing range, and metabolic throughput—not solely through the reproduction maturity gate.